

CODE

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import pygame
import heapq

# ----- SETTINGS -----
WIDTH = 500
HEIGHT = 650
ROWS = 10
CELL = WIDTH // ROWS

# Colors
BG = (10, 10, 30)
GRID = (40, 40, 80)
NEON_GREEN = (57, 255, 20)
NEON_RED = (255, 20, 147)
NEON_BLUE = (0, 200, 255)
NEON_YELLOW = (255, 255, 0)
WHITE = (240, 240, 240)
UI_BG = (20, 20, 50)
pygame.init()
win = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Neon Treasure Hunt")
font = pygame.font.SysFont("Consolas", 24)

# ----- NODE -----
class Node:
    def __init__(self, r, c):
        self.r = r
        self.c = c
        self.color = BG
    def draw(self):
        pygame.draw.rect(win, self.color, (self.c*CELL, self.r*CELL, CELL,
CELL))
        pygame.draw.rect(win, GRID, (self.c*CELL, self.r*CELL, CELL, CELL), 1)
    def pos(self):
        return (self.r, self.c)

# ----- GRID -----
def make_grid():
    return [[Node(i,j) for j in range(ROWS)] for i in range(ROWS)]

def draw(grid, msg="", steps=0):
    win.fill(BG)

    for row in grid:
        for node in row:
            node.draw()
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# UI Panel
pygame.draw.rect(win, UI_BG, (0, 500, WIDTH, 150))

# Buttons
pygame.draw.rect(win, NEON_BLUE, (50, 520, 150, 50))
pygame.draw.rect(win, NEON_RED, (300, 520, 150, 50))
win.blit(font.render("START", True, BG), (85, 530))
win.blit(font.render("RESET", True, BG), (335, 530))

# Message
if msg:
    color = NEON_GREEN if "GOAL" in msg else WHITE
    text = font.render(msg, True, color)
    win.blit(text, (WIDTH//2 - text.get_width()//2, 580))

# Steps
if steps > 0:
    step_text = font.render(f"Steps: {steps}", True, NEON_YELLOW)
    win.blit(step_text, (WIDTH//2 - step_text.get_width()//2, 610))
pygame.display.update()

# ----- A* -----
def heuristic(a,b):
    return abs(a[0]-b[0]) + abs(a[1]-b[1])

def get_neighbors(node, grid):
    res=[]
    r,c=node.pos()
    for dx,dy in [(0,1),(1,0),(0,-1),(-1,0)]:
        nr,nc=r+dx,c+dy
        if 0<=nr<ROWS and 0<=nc<ROWS:
            if grid[nr][nc].color != NEON_BLUE:
                res.append(grid[nr][nc])
    return res

def astar(grid, start, end):
    open_list=[]
    count=0
    heapq.heappush(open_list,(0,count,start))

    came={}
    g={start:0}
    while open_list:
        _,_,current=heapq.heappop(open_list)
        if current==end:
            path=[]
            while current in came:

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        path.append(current)
        current=came[current]
    return path[::-1]

    for n in get_neighbors(current,grid):
        temp=g[current]+1
        if n not in g or temp<g[n]:
            g[n]=temp
            f=temp+heuristic(n.pos(),end.pos())
            count+=1
            heapq.heappush(open_list,(f,count,n))
            came[n]=current

    return None

def main():
    grid=make_grid()
    start=None
    end=None
    msg=""
    steps=0
    run=True
    while run:
        draw(grid,msg,steps)
        for e in pygame.event.get():
            if e.type==pygame.QUIT:
                run=False

        # LEFT CLICK
        if e.type == pygame.MOUSEBUTTONDOWN and e.button == 1:
            x,y = pygame.mouse.get_pos()

            # START button
            if 50<=x<=200 and 520<=y<=570 and start and end:
                msg="Searching..."
                steps=0
                path=astar(grid,start,end)
                if path:
                    for n in path:
                        if n!=start and n!=end:
                            n.color=NEON_YELLOW
                            steps+=1
                    draw(grid,msg,steps)
                    pygame.time.delay(70)
                    msg=" GOAL REACHED!"
                else:
                    msg=" No Path Found"

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        # RESET button
        elif 300<=x<=450 and 520<=y<=570:
            grid=make_grid()
            start=None
            end=None
            msg=""
            steps=0

        # GRID CLICK
        elif y < WIDTH:
            r,c=y//CELL,x//CELL
            node=grid[r][c]
            if not start:
                start=node
                node.color=NEON_GREEN
            elif not end:
                end=node
                node.color=NEON_RED
            else:
                node.color=NEON_BLUE

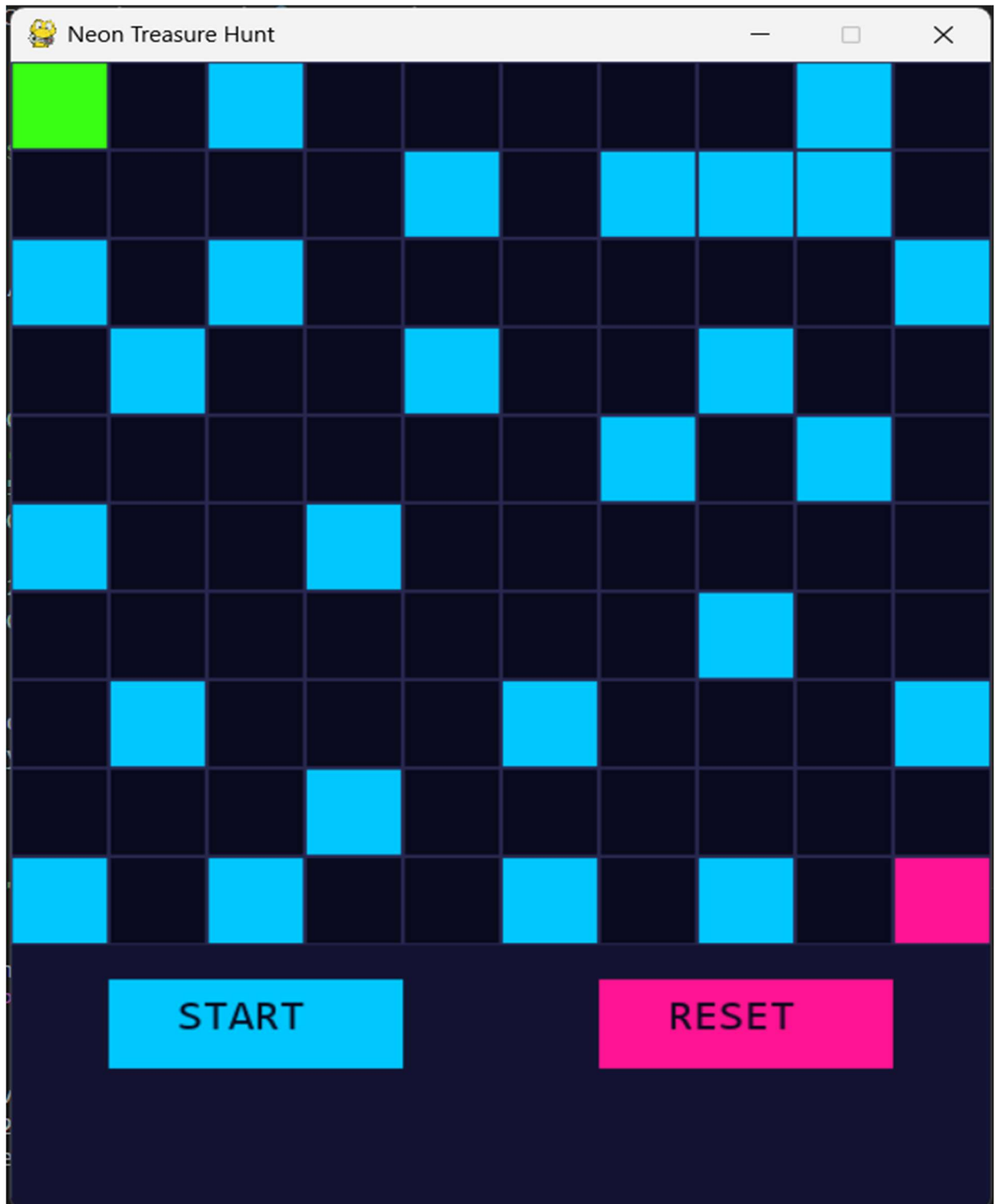
        # RIGHT CLICK
        if e.type == pygame.MOUSEBUTTONDOWN and e.button == 3:
            x,y=pygame.mouse.get_pos()
            if y < WIDTH:
                r,c=y//CELL,x//CELL
                node=grid[r][c]
                node.color=BG
                if node==start: start=None
                if node==end: end=None

    pygame.quit()

# ----- RUN -----
if __name__ == "__main__":
    print("Game is starting...")
    main()

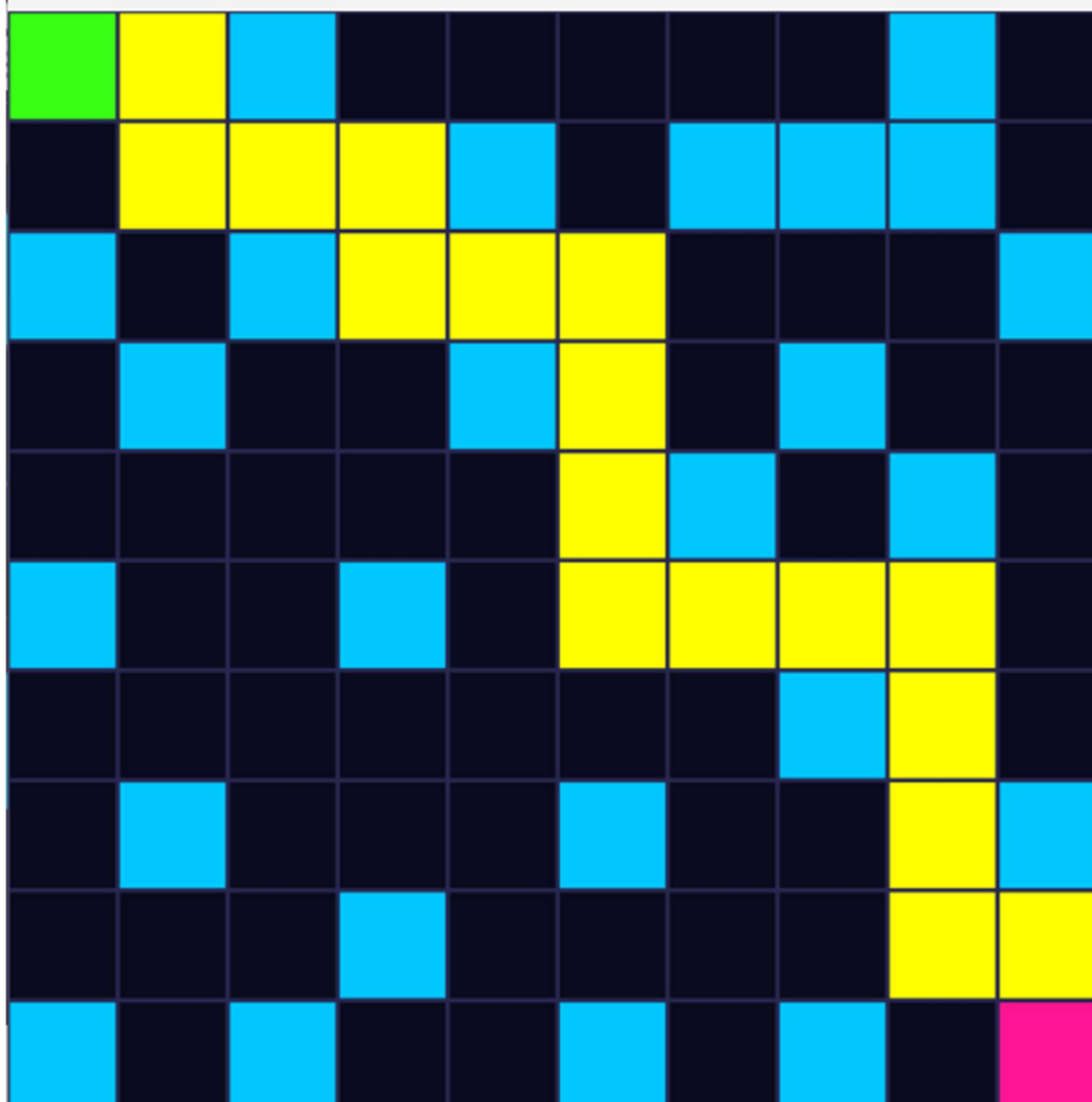
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Output





Neon Treasure Hunt



START

RESET

GOAL REACHED!

Steps: 17